MINISTRY OF EDUCATION AND TRAINING

UNIVERSITY OF ECONOMICS HO CHI MINH CITY

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UEH
UNIVERSITY

FINAL REPORT

COURSE: INTRODUCTION TO HUMAN-COMPUTER INTERACTION

Lecturer: Lê Khánh Duy

Group Members:

Hoàng Lưu Hoài Bảo - 31231023697

Nguyễn Võ Hoàng Nhật - 31231025613

Đỗ Lê Phúc Đức - 31231020173

Trần Trọng Nguyễn - 31231024329

Nguyễn Hà My - 31231027835

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CHAPTER 1: INTRODUCTION

1.1. Motivations

The library is an essential part of the academic environment for university students. As a place that provides knowledge and supports research, it needs to be continuously improved to better meet the increasingly diverse needs of its users. In the context of digital transformation and the growing application of technology in education, building an efficient, user-friendly, and accessible book borrowing system has become more necessary than ever.

Recognizing the importance of the library, our team aims to contribute to enhancing the user experience for UEH students, especially new users or those unfamiliar with the book search and borrowing process. By applying knowledge from the Human-Computer Interaction (HCI) course and adopting a user-centered design approach, we strive to create practical solutions that help students access learning materials more conveniently and effectively.

1.2. Problems

Through interviews, observations, and personal experience, our team has identified several problems that students face when borrowing books at the UEH library. First, students have difficulty finding the books they need because the books are not well organized on the shelves. Often, they are unsure about the exact location of the book they are looking for. Additionally, many books are placed in the "quiet study" area, which requires silence. This makes it uncomfortable for students to search for books, as it may disturb others who are studying. As a result, students feel shy and less likely to actively look for the books they need. Furthermore, the self-service book borrowing machines at the library have several issues. The interface is not user-friendly, making it difficult for new users to navigate, and the system often experiences lag or delays. These problems not only cause inconvenience but also affect students' ability to access study materials and disrupt their learning experience.

→ Therefore, the team has decided to find solutions to improve the book borrowing process, enhance the user experience, and create a more comfortable and modern learning environment for UEH students.

1.3. Objective

Our objective is to design and develop solutions that fully or partially address the difficulties users currently face when searching for and borrowing books at the

University of Economics Ho Chi Minh City (UEH) library. These challenges range from unclear book placement and an unfriendly search interface to difficulties in using the self-service borrowing machines or accessing books on shelves that are too high. The proposed solutions aim not only to improve the efficiency of the book searching and borrowing process but also to enhance the overall user experience, making the library more convenient, accessible, and modern. The new system is expected to meet the needs of at least two main user groups: UEH students, who regularly use the library for studying and research, and first-time users, including freshmen, visitors, or those unfamiliar with how the library operates. Ultimately, the goal of the project is to build a more friendly, modern, and user-centered learning environment that better suits the habits and needs of UEH library users.

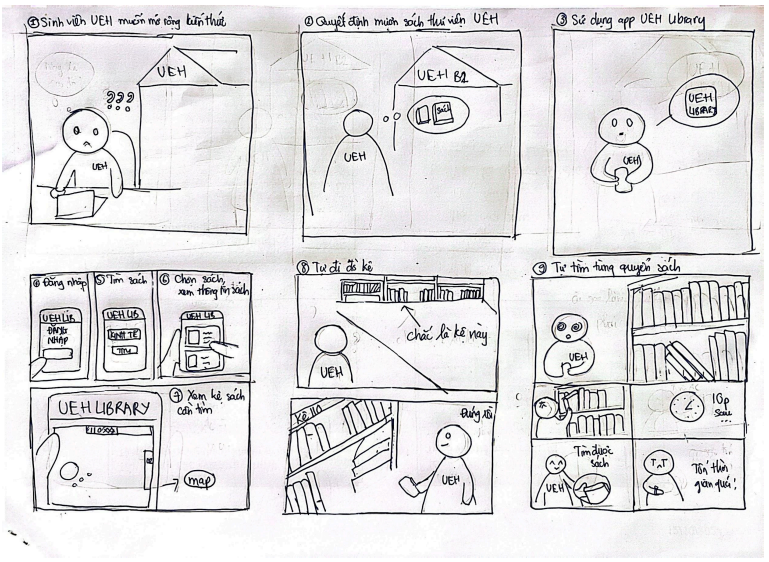
CHAPTER 2: PRIMITIVE PROTOTYPE

2.1. Solution 1:

2.1.1. Motivation and solution

In the traditional way of borrowing books, students often spend a lot of time to determine the exact location of the book, especially in a space with many floors and many rows of shelves. Finding books mainly depends on the classification code or asking the librarian. This wastes time and reduces learning efficiency. We propose a 2D map interface that shows the library structure from a top-down view. When a user searches for a book, the system displays the exact location of the shelf containing that book on the map, while marking the location and name of the shelf.

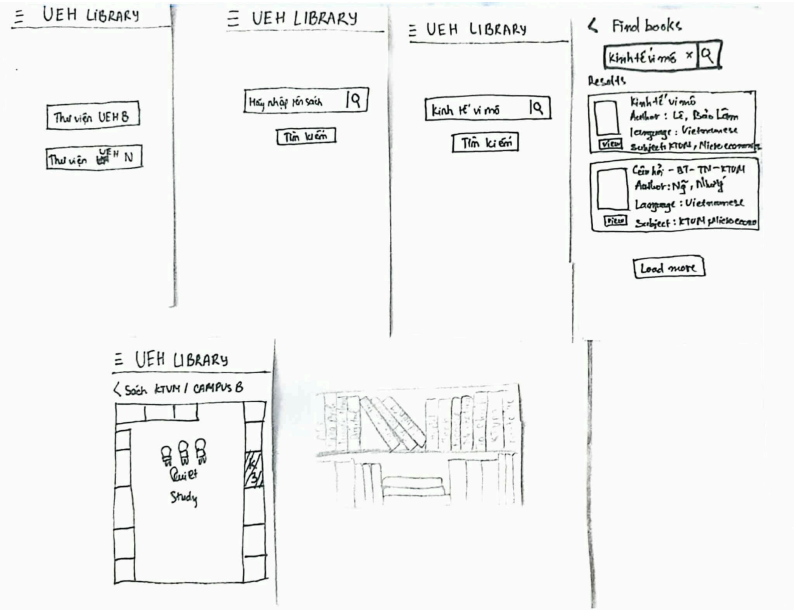
2.1.2. Storyboard



The storyboard outlines the user journey as follows:

+ **Step 1:** Recognize learning needs
+ **Step 2:** Decide to borrow books
+ **Step 3:** Use UEH Library application
+ **Step 4:** Log in
+ **Step 5:** Search for books
+ **Step 6:** View book information
+ **Step 7:** View library map
+ **Step 8:** Find book shelves
+ **Step 9:** Find book & borrow successfully

2.1.3. Paper prototype



+ **Step 1:** Select the UEH library you want to search
+ **Step 2:** Enter the keyword of the book you want to search
+ **Step 3:** Click on the search button
+ **Step 4:** View the list of book results
+ **Step 5:** Select the book which you want to borrow to view details
+ **Step 6:** View the map of the bookshelf location in the library
+ **Step 7:** Go to the correct area and shelf to get the book

2.1.4. Strengths

- **Intuitive:** 2D map helps students visualize the library space easily.
- **Accessible:** No high-end equipment required, suitable for all types of phones.
- **User-friendly:** Simple interface, easy to use even for first-time users.

- **Feasible implementation:** Suitable for current library reality if there is already a space diagram

2.1.5. Weaknesses

- **Difficult to use in complex libraries:** For libraries with asymmetrical structures or multiple, intermittent levels, 2D maps can be confusing.
- **Lack of interactivity:** Users cannot rotate the map or find their way around in real time like in Google Maps.
- **No support for the visually impaired:** The interface is only images, no voice integration or special accessibility support.

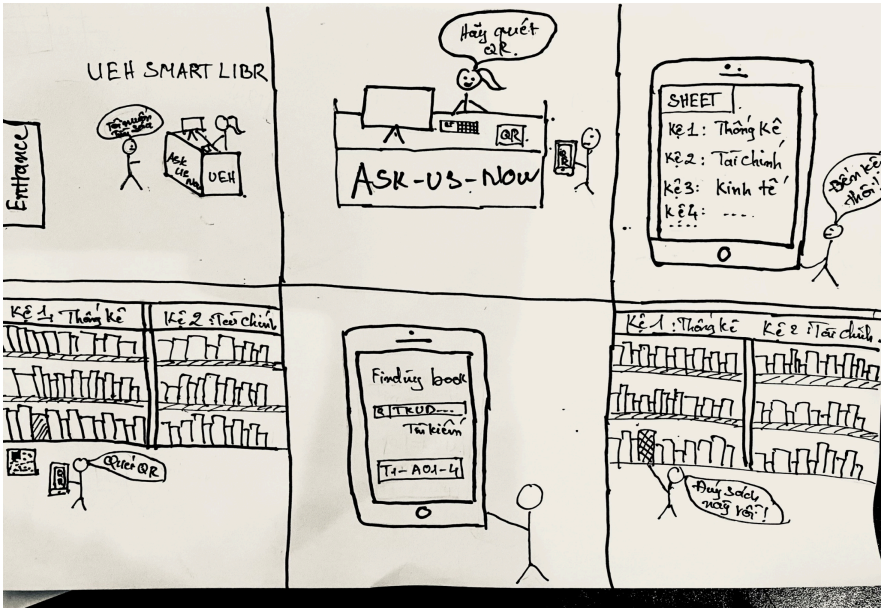
2.2. Solution 2:

2.2.1. Motivation and solution

In modern libraries, finding a book efficiently can be challenging, especially in large institutions with numerous shelves. The main motivation behind this solution is to reduce the time and effort required to locate a book within the library. By using QR scanning technology, we enable users to quickly access information about the book's location and receive guided instructions to the designated shelf.

→ **Proposed Solution:** To address this problem, we propose a **QR Scan Book Finder System**. This system will simplify the process of finding books by providing a structured, step-by-step method using QR code scanning.

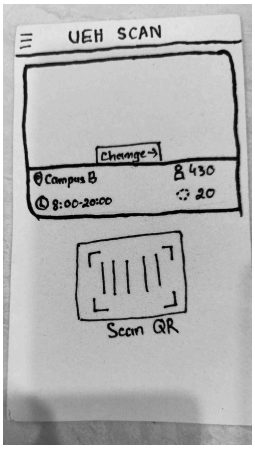
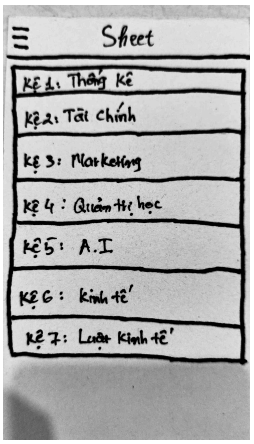
2.2.2. Storyboard

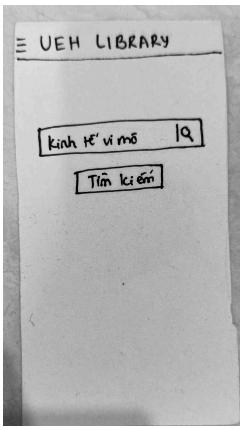
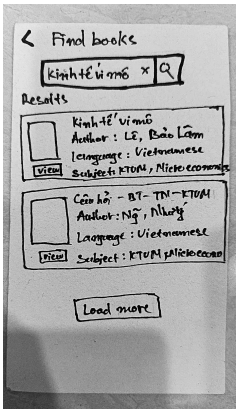


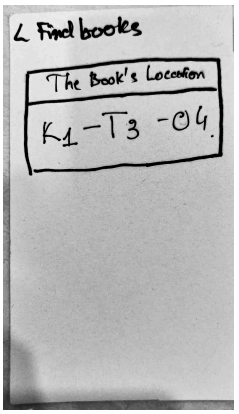
The storyboard outlines the user journey as follows:

+ **Step 1:** Entering the Library.
+ **Step 2:** Following the Map.
+ **Step 3:** Arriving at the Bookshelf.
+ **Step 4:** Finding the Book.

2.2.3. Paper prototype

	**Step 1:** Login and Scan QR - **Process:** + The user enters the library and scans the QR code at the entrance. + The system displays a **digital sheet of bookshelves** categorized by subjects like Economics, Computer Science, Finance, etc. - **Purpose:** + Efficiently log the user in and display the categorized list of bookshelves for easy navigation.
	**Step 2:** Select Bookshelf - **Process:** + After scanning, the system shows the categorized bookshelf list. + The list includes subjects such as **AI, Economics, Marketing, Management, Finance, and more**. - **Purpose:** + Allow users to choose the correct subject area for the book they are looking for.

	**Step 3:** Search for Book - **Process:** + The user types the book name (e.g., **"Kinh tế vĩ mô"**) and clicks the **"Search"** button. + The system displays the search results, including book title, author, and availability. - **Purpose:** + Quickly find relevant books without manually searching the entire library.
	**Step 4:** Book Information Display - **Process:** + The system shows the detailed search results, such as: - **Title:** Kinh tế vĩ mô - **Author:** Lê Bảo Lâm - **Language:** Vietnamese - **Subject:** Microeconomics - **Purpose:** + Provide accurate and detailed information about the book before guiding the user to the location.

	**Step 5:** Book Location - **Process:** + The system provides the precise location of the book on the shelf, for example: **K1 - T3 - 04** - **Purpose:** + Minimize confusion by giving an exact position on the shelf, reducing search time.

2.2.4. Strengths

- **Efficiency:** Reduces search time by providing a clear path to the book's location.
- **User-friendly:** Simple QR scan mechanism to access the bookshelf layout and book details.
- **Scalability:** Adaptable to various library layouts and book arrangements.
- **Integration:** Can be combined with existing library management systems.

2.2.5. Weaknesses

- **Dependency on QR Codes:** Users without a QR-scanning device may face difficulties.
- **Navigation Challenges:** Users unfamiliar with the library layout may still face challenges despite the map.
- **Network Issues:** Requires a stable internet connection for real-time updates.
- **Technical Maintenance:** Needs regular updates to ensure QR codes are accurate and functional.

2.3. Solution 3:

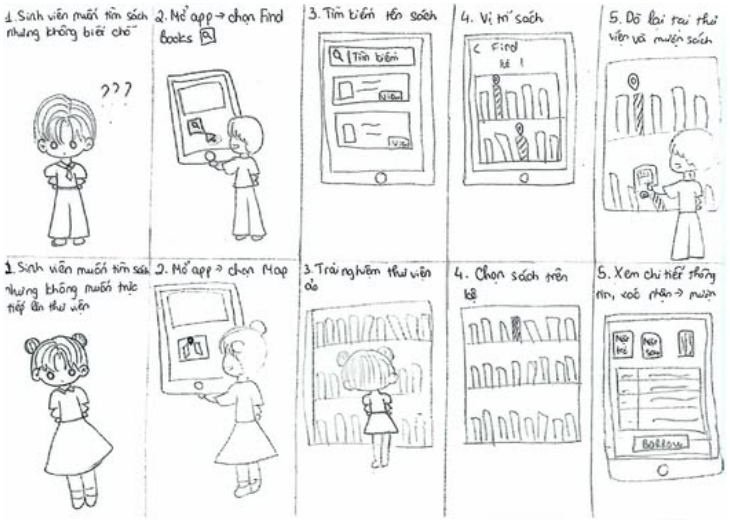
2.3.1. Motivation and solution

In the context of rapid digital transformation and increasingly flexible learning demands, UEH students now expect more than just a traditional library — they seek a smart, convenient, and time-saving learning environment. However, navigating a large, multi-level library with subject-specific zones remains a real challenge, especially for new students or those unfamiliar with the layout. Furthermore, some

introverted students may feel hesitant to wander through study areas just to locate a single book, or may feel uncomfortable asking for assistance. Even when they manage to find the book, it might be placed too high or low to access easily — adding to the frustration.

→ These real-life difficulties have motivated us to develop a comprehensive solution that leverages technology to improve the library experience at UEH. As technology advances, so do user expectations — they no longer just want to "find the book," but also expect a seamless, personalized, and user-friendly journey within the library.

2.3.2. Storyboard



The storyboard outlines 2 user journey as follows:

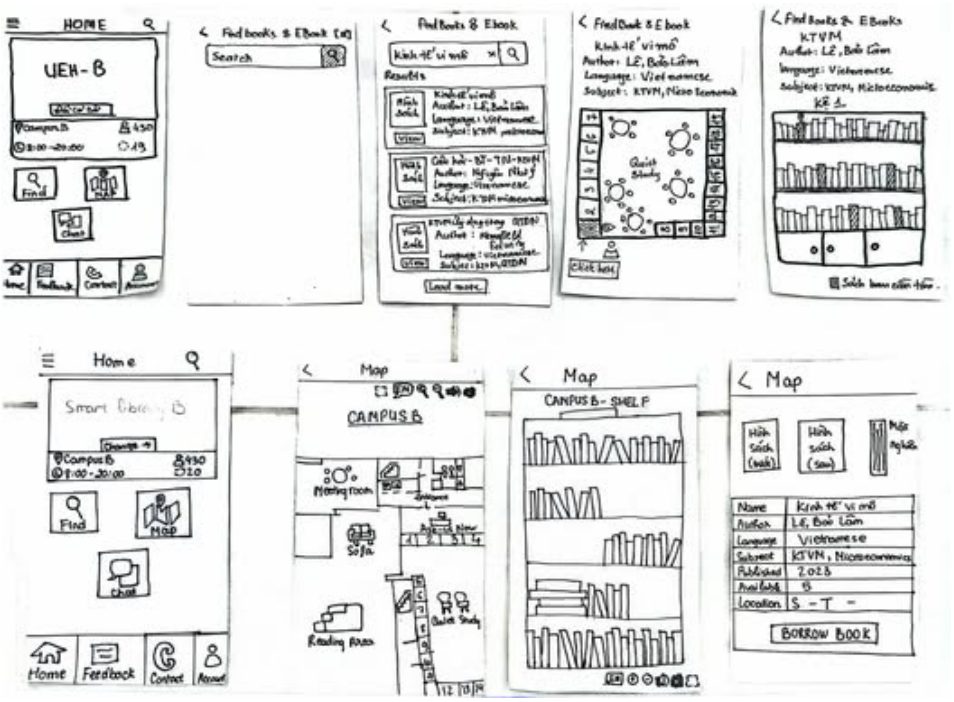
→ First Journey:

+ **Step 1:** Student does not know the location of the book.
+ **Step 2:** Open the application and choose the “Find Books” function.
+ **Step 3:** Type the name of the book.
+ **Step 4:** See the location of the book.
+ **Step 5:** Go to the exact location of the book and borrow it.

→ Second Journey:

+ **Step 1:** Student wants to find books but does not want to go to the library.
+ **Step 2:** Open the application and choose the “Find Books” function.
+ **Step 3:** Experience the virtual library on the application.
+ **Step 4:** Choose the suitable book being placed on the shelf.
+ **Step 5:** View the details of that book and confirm to borrow it.

2.3.3. Paper prototype



+ **Step 1:** Choose the function that you want to use.
+ **Step 2:** Type the name of the book that you want to find.
+ **Step 3:** Choose the book that you want to find from the result of searching.
+ **Step 4:** View the details of that book and the location of the shelf where that book is placed.
+ **Step 5:** Click on the correct shelf and view the location of that book being placed on the shelf.
+ **Step 6:** Go back to the dashboard and choose the map function.
+ **Step 7:** View the 2D map of the library, you can click on the quiet study area and choose the appropriate shelf.
+ **Step 8:** View all the books which are placed on that shelf.
+ **Step 9:** Click on any books to fully view the details of that book.

2.3.4. Strengths

- A unique and engaging feature that enhances the library experience
- Visual display of book covers to help users quickly recognize titles
- Real-time display of book availability and clear, intuitive location tracking
- Comprehensive book information, including visuals and content summaries
- User-friendly 3D map that makes it easy to locate shelves and navigate the space

2.3.5. Weaknesses

- The interface requires refinement in terms of button placement and interaction design to improve usability.
- An overload of features currently creates a cluttered experience, making it difficult for users to navigate smoothly.
- There is a lack of clear action cues or on-screen prompts to help users understand what interactions are expected at each step.
- The detection mechanism remains vague — it is not clearly explained how it functions or whether it supports real-time operation.

2.4. Reason for chosen solution

We propose adopting the prototype 3 instead of the previous 2D map and QR code scanning version because it offers a much richer and more intuitive experience. Unlike a flat 2D diagram, the virtual 3D map allows users to "walk" through a simulated library environment—enabling them to navigate directly to specific aisles and shelves. Compared to the QR code method, which required students to visit the library counter and scan each tag individually, this solution supports remote access: users can preview book covers, check availability, and locate shelf positions simply by entering keywords. Moreover, features such as real-time updates, comprehensive book information (including images and summaries), and clear visual cues significantly reduce confusion about book locations. This user-centered design not only speeds up the search process but also minimizes physical movement and cognitive load, making book-finding faster, more accurate, and more pleasant.

In addition, the modular architecture of this prototype allows seamless integration with the current library management system and future scalability. By sharing the same real-time data used by staff for shelving, it ensures that both students and library personnel operate with the most accurate and consistent information. This unified approach not only reduces redundant tasks—staff no longer need to repeatedly answer **"Where is this book?"**—but also enables borrowing trend analysis, helping the library optimize shelf layouts based on actual demand.

CHAPTER 3: HIGH-FIDELITY DESIGN DECISIONS

3.1. User Interface

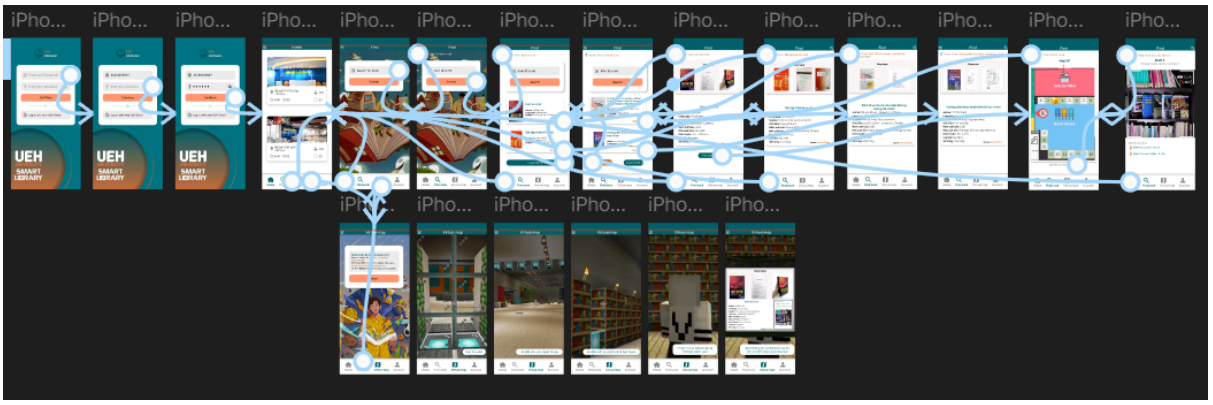
The high-fidelity (hi-fi) prototype can be viewed through the demo video here, and the interactive version of the application is also available. The hi-fi interface was designed to enhance the book-borrowing experience by integrating essential features

into the “UEH Smart Library” application. This prototype simulates the complete user journey and offers an intuitive experience—from launching the app, searching for books, to exploring the virtual library.

A total of 22 screens were created to reflect the full user experience, with a focus on the two main features of the app: Book Finding and Virtual Tour. Upon launching the app, users are taken to the login screen. Here, they can either sign in with their UEH account or continue as a guest. These initial screens are designed with simplicity and accessibility in mind, making them user-friendly for all audiences. The app interface uses a fixed bottom navigation bar, allowing users to easily switch between the main sections: Home, Find, Virtual Tour, and Profile. All screens are logically connected, ensuring a smooth and seamless interaction flow.

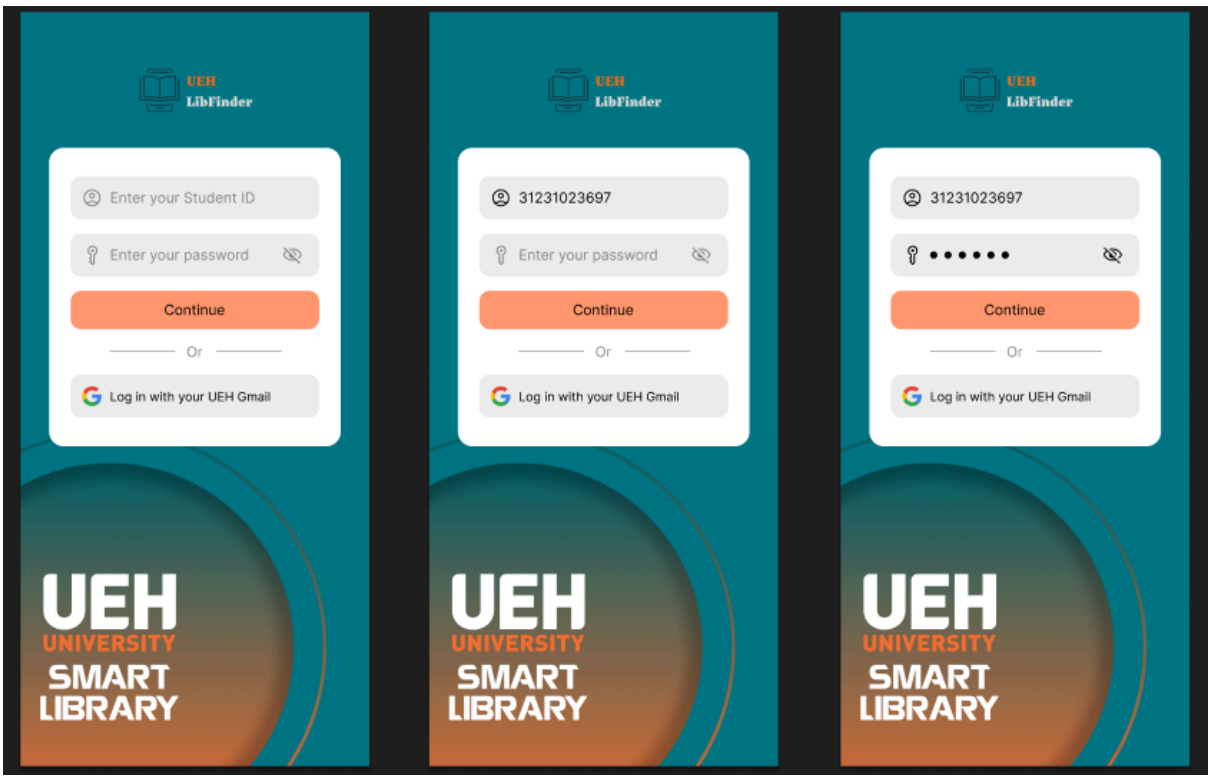
3.2. Core Features

The hi-fi prototype presents a smooth and intuitive interaction flow. Users start from the login screen and navigate through main features via a fixed bottom menu: Home, Find, Virtual Tour, and Profile. In the **Find** flow, users search for books, view results, and follow a 3D map to locate books on shelves. In the **Virtual Tour**, users explore the library in first-person view and tap on books to access details. All screens are logically connected to support seamless navigation and interaction.

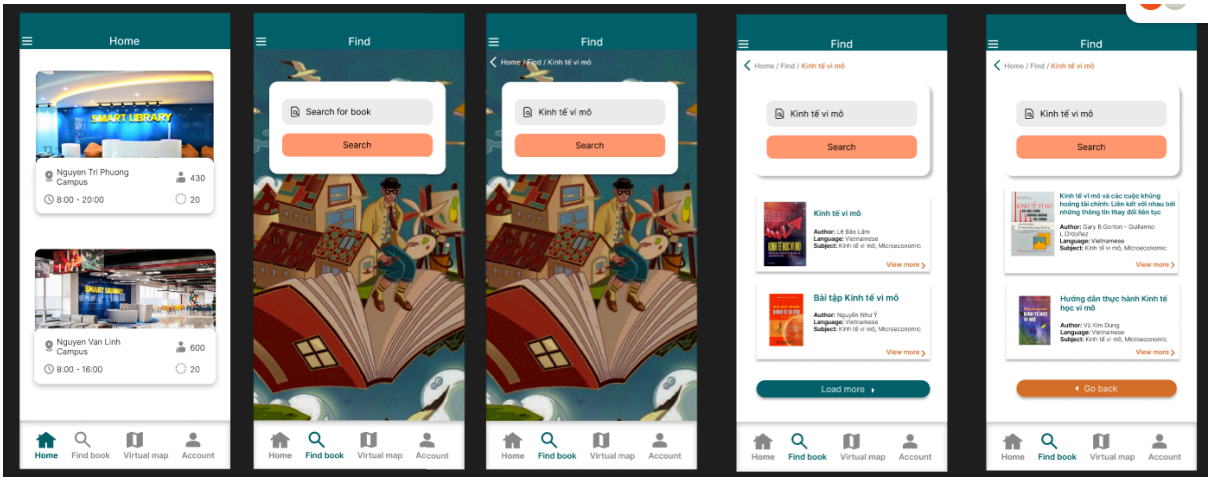


3.2.1. Book Search

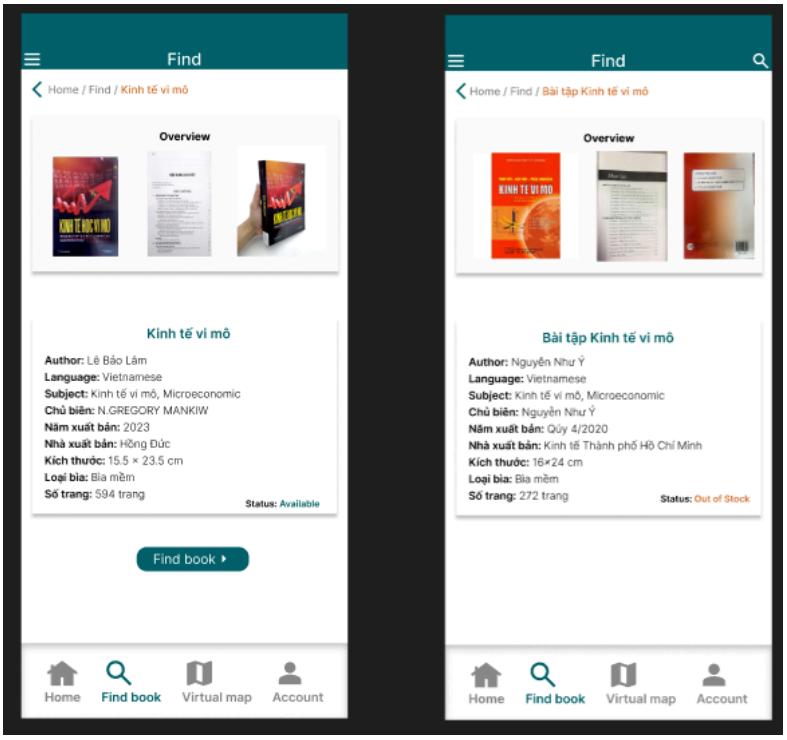
Upon opening the app, users are taken to the login screen. They can log in using a UEH account or continue as a guest. These initial screens are designed with simplicity and accessibility in mind, making them easy to use for all types of users.



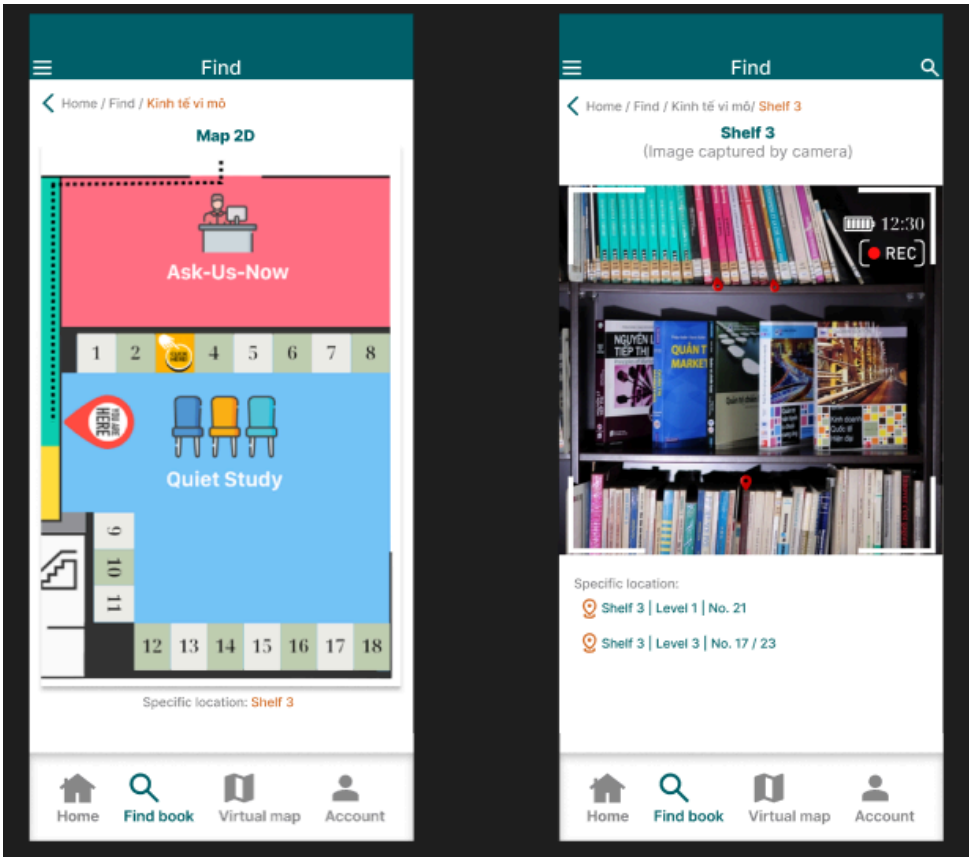
The first key feature is **Book Finding**. From the bottom navigation bar, users can access the search function and enter a book title or any keyword. The app then displays a list of matching results, showing the book cover, title, and author. A standout feature here is the integration of a **3D library map**, which helps users locate the exact position of the book within the library. Once a book is selected, the app directs users to a 3D map view of the library, with the book's location clearly marked on the shelves. Users can view the overall layout, including functional areas and aisles between shelves, making it easier to navigate and reach the correct shelf.



If the book is unavailable (e.g., already borrowed), the app displays a notice indicating the book is not currently available, and the "Find Book" button will be hidden.

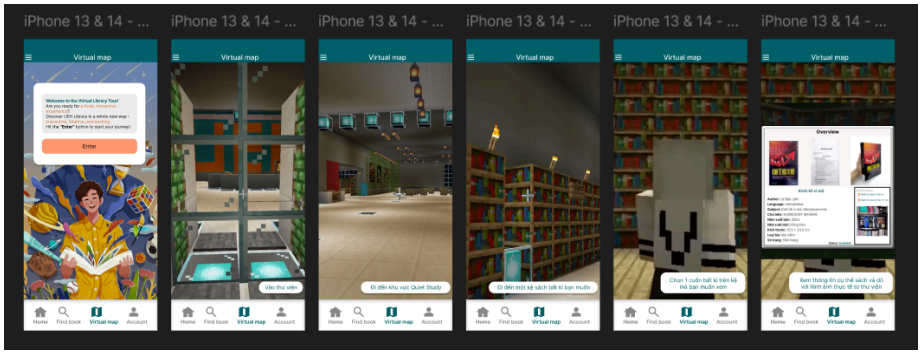


After navigating to the book's location, users can access detailed information about the book. This includes the title, author, a description, the number of copies left, the shelf location, and related book suggestions. The layout is clean and reading-focused, helping users decide whether to borrow the book or explore further resources.



3.2.2. Virtual Library

In addition, the **Virtual Tour** function offers a unique way to explore the library. Users can virtually walk through the library in a **first-person 3D environment**. The virtual space is modeled in detail, including areas such as hallways, staircases (specific to UEH Library N1), reading zones, group study rooms, private study spaces, and bookshelves. Users can freely move around using navigation buttons or swipe gestures, simulating the experience of walking through a real library. While exploring, users can approach any shelf and tap on visible books to view detailed information—just like in the Book Finding function. This feature creates a highly interactive experience that blends exploration with learning.



CHAPTER 4: EVALUATION

4.1. Scope

- **Total participants:** Up to 4 university students.
- **Brief profile:** Aged between 18 and 22, frequent library users for studying and research, with moderate to high digital proficiency.
- **Reason for selection:** This group represents the target users of the project. They typically need to locate books or reference materials related to their academic courses in the library while preferring to minimize social interactions. This aligns well with the prototype's features that support autonomous and discreet book searching.

User target	Age	Gender	Major	Library usage frequency	Tech proficiency	Book searching habit	Comfort with social interaction
A	19	Male	Logistics	1-2 times per week	Good	Once a week	Shy
B	19	Male	Technology and Innovation	Three times a week	Very good	Twice a week	Quite shy
C	20	Female	Marketing	Four times a week	Good	Once a week	Introvert
D	19	Male	Statistics	Twice a week	Good	1-2 time per week	a bit shy

4.2. Evaluation Method

4.2.1. Independent variables to measure

+ **System:** Proposed system
+ **Target users (by age):** Students
+ **Tasks:**

    + Search for a book using keywords, then find the book based on the image and specific location provided by the system.
    + Experience the virtual map, and use this feature to select a book without having to go to the library physically.

→ The tasks are simple enough that counterbalancing the order is unnecessary.

+ **Order of testing and prior experience:**
    + We only have 2 conditions (1 system × 2 tasks), so the team employs a **within - subject method.**

4.2.2. Quantitative measures to measure

To evaluate the effectiveness and user experience of the prototype, our team defined a set of performance and perception metrics aligned with the core principles of Human-Computer Interaction (HCI). The metrics applied in this evaluation include:

+ **Usability, Learnability**
+ **Efficiency**
+ **Satisfaction**
+ **Mental Model**
+ **Visibility & Feedback**
+ **Error Prevention & Recovery**
+ **Accessibility**
+ **General Usability & Acceptance**

4.2.3. Content

Although ideally users should evaluate both systems on a 1–5 scale, this would take too much time since the interviewer requires the process to be brief. Therefore, the team decided to compress the evaluation of both systems into a single 1 to 5 scale, where:

+ **1 = the old system is better,**
+ **5 = the proposed new system is better, and**
+ **3 = both systems are equivalent.**

4.2.4. Data Collection

After the user completes the form, the team also asks them a few additional questions about their experience right after using the system, as well as any suggestions they might have for improving it (if any). These follow-up questions also

focus on HCI (Human-Computer Interaction) aspects. Some of the collected questions include:

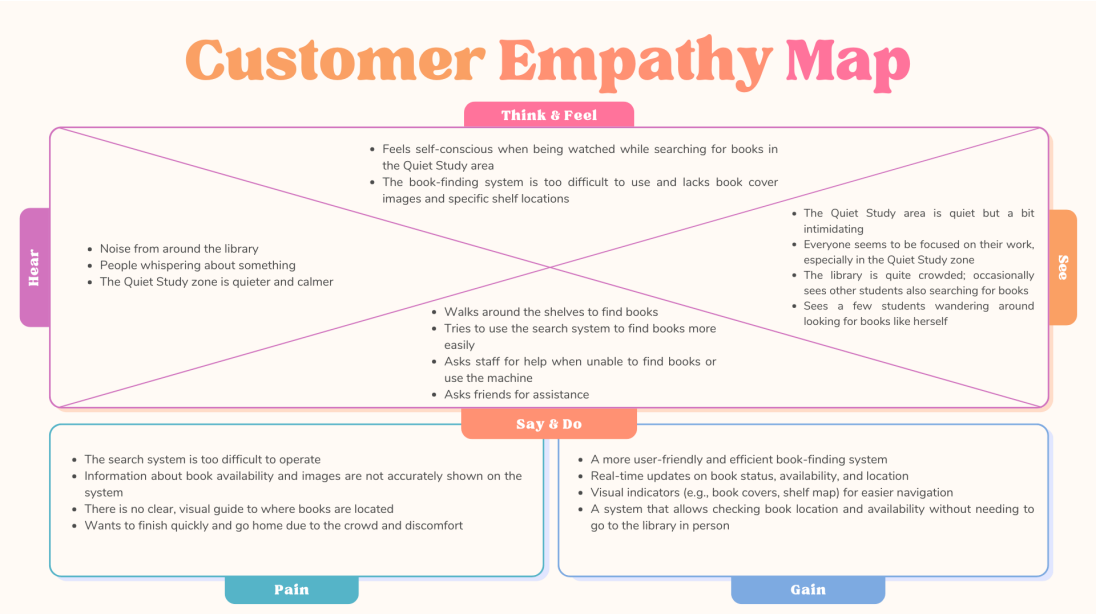
+ Did you find the interface and interactions easy to use?
+ Did you understand how to use the system from the very first time?
+ Did the system help you find books faster compared to the usual way?
+ How did you feel after using the system? Were you satisfied?
+ Did the system provide clear feedback after each action?
+ If you had low vision or used a low-performance device, was the system still accessible?

4.3. Participants

4.3.1. Persona & Empathy



Hình Persona



Hình Empathy map

4.3.2. Lo-fi testing: Link

During PA3, the team created three paper-based low-fidelity prototypes:

- A 2D library map.
- A QR-based book-finding system.
- A smart search system with a 3D virtual map.

→ Users were guided through simulated scenarios to perform tasks such as scanning codes, navigating sections, and accessing map features. Most feedback emphasized clarity, ease of use, and the ability to locate books efficiently.

4.3.3. Hi-fi testing: Link

In PA4, a high-fidelity prototype was developed using Figma to simulate the Smart Library system. Testing was conducted at UEH Campus B with 2 in-person and 3 remote participants. Tasks included:

- Searching for books using keywords.
- Viewing the map for the book's location
- Exploring the virtual navigation system.

→ Results: Most users completed tasks successfully within 45–90 seconds and described the system as "intuitive" and "less confusing" compared to traditional methods.

4.4. Team responsibilities

4.4.1. During problem identification

- **Duc:** On-site observations.
- **Nhat:** Self-immersion testing.
- **Nguyen:** Flow & sequence modeling.
- **My:** Persona development.
- **Bao:** Empathy map & affinity diagram synthesis.

4.4.2. Lo-fi testing

- **Bao & Nhat:** Paper prototype creation.
- **Duc:** Scenario preparation.
- **Nguyen:** User walkthroughs, paper prototype creation (prototype 2).
- **My:** Note-taking and feedback collection.

4.4.3. Hi-fi testing

- **Bao:** interface design on figma, Post-test interviews.
- **Nguyen:** design minecraft 3D space (3D map), Post-test interviews.
- **Duc:** design minecraft 3D space (3D map), Post-test interviews.
- **Nhat:** design minecraft 3D space (3D map), Post-test interviews.
- **My:** Data consolidation from response forms.

4.5. Feedback Collection

- **Problem identification:** Link
- **Lo-fi testing:** Link
- **Hi-fi testing:** Link

4.6. Session Duration

- **Lo-fi Testing:** ~10 minutes per session.
- **Hi-fi Testing:** 15–20 minutes per session, consisting of:
    + 5 minutes: Introduction & setup.
    + 7–10 minutes: User interaction.
    + 5 minutes: Interview and reflections.

4.7. Measures & Metrics

4.7.1. During problem identification

To ensure that our solution addresses real user needs, we incorporated early-stage metrics during the problem identification phase:

- **Frequency of Pain Points:** Through interviews and observations, we recorded how often users encountered issues such as “difficulty finding books” or “hesitation to ask staff.”
- **Emotional Indicators:** Users expressed discomfort in quiet zones (e.g., “I don’t want to disturb others”), which served as qualitative signals of system breakdowns.
- **User Quotes & Insights:** We used structured interviews to extract first-principle insights about root frustrations rather than surface-level complaints. For instance, users did not merely dislike “lack of guidance”—they fundamentally wanted to act independently in a social setting without judgment.
- **Affinity Mapping:** We grouped user needs into categories aligned with HCI principles (e.g., mental model mismatch, visibility issues) to guide design priorities.

4.7.2. Lo-fi testing

Lo-fi testing allowed us to evaluate the clarity of the system structure and interaction flow using paper prototypes and early sketches:

- **Task Completion Observation:** Users were given tasks (e.g., find a book on the 3rd floor). We observed if they understood the system logic without prompting.
- **Navigation Clarity:** We evaluated whether users could navigate the 2D map or flowchart without external help.
- **Mental Model Alignment:** We noted moments of hesitation or confusion, indicating mismatches between users’ expectations and the system's logic.
- **Feedback Style:** Observers captured real-time reactions such as “this icon is confusing” or “where do I click next?”—used later in the affinity diagram.

4.7.3. Hi-fi testing



Phương diện HCI	Câu hỏi phỏng vấn gợi ý	Người dùng - Khánh	Người dùng 2	Người dùng 3	Người dùng - Khôi
Usability (Tính khả dụng)	Bạn có thấy giao diện và thao tác dễ sử dụng không?	Dễ sử dụng	dễ sử dụng dễ hiểu	có, dễ hiểu mà	Dễ sử dụng
Learnability (Dễ học)	Bạn có hiểu cách sử dụng hệ thống ngay từ lần đầu không?	Dễ hiểu, nếu tự mò thì không quá 3 phút	dễ hiểu, có thể tự mò dễ dàng	dễ hiểu,	có, nhìn là biết mà, giới trẻ mà
Efficiency (Hiệu quả thao tác)	Hệ thống có giúp bạn tìm sách nhanh hơn so với thực tế không?	Nhanh, tiện, có thể xem ở nhà trước	có thể xem trước ở nhà, nhan và tiện	Tìm sách nhanh hơn cách truyền thống, tiện, có thể xem trước ở nhà	Đúng ời
Satisfaction (Mức độ hài lòng)	Bạn cảm thấy thế nào sau khi sử dụng hệ thống? Có hài lòng không?	trên thang điểm 10 thì là 9.5	cho các bạn 8.5đ trên 10	Hài lòng 8-9/10	9.5
Mental Model (Mô hình tư duy)	Luồng thao tác có đúng như bạn mong đợi không? Có bị rối không? *Ví dụ 1 (Luồng thao tác hợp lý, đúng kỳ vọng): "Mình nghĩ là dùng như mình hình dung. Sau khi tìm sách thì thấy luôn bản đồ hiển thị vị trí – rất hợp lý. Mình không cần mất thời gian đoán bước tiếp theo."*	luồng thao tác hợp lý	không hoàn toàn giống như mong đợi nhưng mà vẫn rất được trau chuốt và streamlined	Có những cái hướng dẫn tiếng việt rõ ràng, dễ hiểu. Đúng như mình hình dung và rất hợp lý. Không cần đoán các bước tiếp theo sẽ là gì	hong rối
Visibility & Feedback (Hiển thị & phản hồi)	Bạn có thấy hệ thống phản hồi rõ ràng sau mỗi thao tác không?	Mượt và dễ dàng thực hiện	rõ ràng và dễ hiểu	rõ ràng sau mỗi thao tác	mượt, dễ xài

In the hi-fi stage, a functionally complete prototype (in Minecraft or Figma) was tested in conditions resembling real-world use:

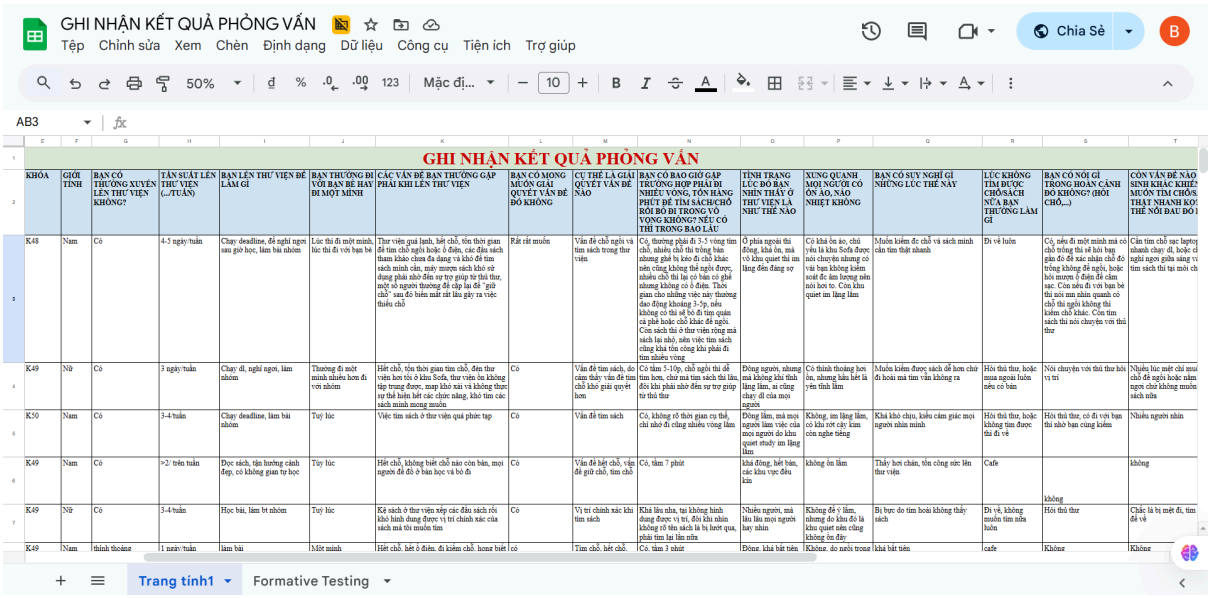
- **Task Completion Rate:** Users were timed and observed while completing tasks like searching for a book and locating it using the virtual map. Most users succeeded without needing help.
- **Time on Task:** Users took 45–70 seconds to complete the assigned tasks, indicating increased efficiency versus traditional systems.
- **Quantitative Scoring (1–5 scale):** Users evaluated the system across 8 HCI aspects—Usability, Learnability, Efficiency, Satisfaction, Mental Model, Visibility & Feedback, Error Prevention & Recovery, and Accessibility. Average scores ranged from 3.63 to 4.75.
- **Post-Test Interview & Survey:** Additional open-ended questions helped capture user satisfaction, emotional responses, and feedback on improvements (e.g., larger text for low-vision users).
- **Error Analysis:** We tracked minor issues (e.g., unclear location labels) and suggested interface fixes to reduce confusion.

→ By applying first principles thinking, our team chose only the most essential, user-centered metrics — allowing us to directly observe how users interact, where they struggle, and how we can refine the system to improve clarity, confidence, and accessibility.

CHAPTER 5: RESULTS

5.1. Interview insights summary

5.1.1. During problem identification



GHI NHẬN KẾT QUẢ PHỎNG VẤN

KHÓA	GIỚI TÍNH	BẠN CÓ THƯỜNG XUYÊN LÊN THƯ VIỆN KHÔNG?	TẦN SUẤT LÊN THƯ VIỆN (../TUẦN)	BẠN LÊN THƯ VIỆN ĐỂ LÀM GÌ	BẠN THƯỜNG ĐI VỚI BẠN BÈ HAY ĐI MỘT MÌNH	CÁC VẤN ĐỀ BẠN THƯỜNG GẶP PHẢI KHI LÊN THƯ VIỆN	BẠN CÓ MONG MUỐN GIẢI QUYẾT VẤN ĐỀ ĐÓ KHÔNG	CỤ THỂ LÀ GIẢI QUYẾT VẤN ĐỀ NÀO	BẠN CÓ BAO GIỜ GẶP TRƯỜNG HỢP PHẢI ĐI NHIỀU VÒNG, TỐN HÀNG PHÚT ĐỂ TÌM SÁCH/CHỜ RỒI BỎ ĐI TRONG VÔ VỌNG KHÔNG? NẾU CÓ THÌ TRONG BAO LÂU	TÌNH TRẠNG LÚC ĐÓ BẠN NHÌN THẤY Ở THƯ VIỆN LÀ NHƯ THẾ NÀO	XUNG QUANH MỌI NGƯỜI CÓ ỒN ÀO, NÁO NHIỆT KHÔNG	BẠN CÓ SUY NGHĨ GÌ NHỮNG LÚC THẾ NÀY	LÚC KHÔNG TÌM ĐƯỢC CHỖ/SÁCH NỮA BẠN THƯỜNG LÀM GÌ	BẠN CÓ NÓI GÌ TRONG HOÀN CẢNH ĐÓ KHÔNG? (HỎI CHỖ,...)	CÒN VẤN ĐỀ NÀO SINH KHÁC KHIẾ MUỐN TÌM CHỖ/S THẬT NHANH KO? THẾ NÓI ĐẦU ĐÓ
K48	Nam	Có	4-5 ngày/tuần	Chạy deadline, để nghỉ ngơi sau giờ học, làm bài nhóm	Lúc thì đi một mình, lúc thì đi với bạn bè	Thư viện quá lạnh, hết chỗ, tốn thời gian để tìm chỗ ngồi hoặc ổ điện, các đầu sách tham khảo chưa đa dạng và khó để tìm sách mình cần, mấy mượn sách khó nữ dụng phải nhờ đến sự trợ giúp từ thủ thư, một nữ người thường để cặp lại để "giữ chỗ" sau đó biến mất rất lâu gây ra việc thiếu chỗ	Rất rất muốn	Vấn đề chỗ ngồi và tìm sách trong thư viện	Có, thường phải đi 3-5 vòng tìm chỗ, nhiều chỗ thì trông bàn nhưng ghế bị kéo đi chỗ khác nên cũng không thể ngồi được, nhiều chỗ thì lại có bàn có ghế nhưng không có ổ điện. Thời gian cho những việc này thường dao động khoảng 3-5p, nếu không có thì sẽ bỏ đi tìm quán cà phê hoặc chỗ khác để ngồi. Còn sách thì ở thư viện rộng mà sách lại nhỏ, nên việc tìm sách cũng khá tốn công khi phải đi tìm nhiều vòng	Ồ phía ngoài thì đông, khá ồn, mà vô khu quiet thì im lặng đến đáng sợ	Có khá ồn ào, chủ yếu là khu Sofa được nói chuyện nhưng có vài bạn không kiểm soát đc âm lượng nên nói hơi to. Còn khu quiet im lặng lắm	Muốn kiếm đc chỗ và sách mình cần tìm thật nhanh	Đi về luôn	Có, nếu đi một mình mà có chỗ trông thì sẽ hỏi bạ gần đó để xác nhận chỗ đó trông không để ngồi, hoặc hỏi mượn ổ điện để cắm sạc. Còn nếu đi với bạn bè thì nói mn nhìn quanh có chỗ thì ngồi không thì kiếm chỗ khác. Còn tìm sách thì nói chuyện với thủ thư	Cần tìm chỗ sạc laptop nhanh chạy đi, hoặc cũ nghỉ ngơi giữa sáng v tìm sách thì tại mỏi ch
K49	Nữ	Có	3 ngày/tuần	Chạy đl, nghỉ ngơi, làm nhóm	Thường đi một mình nhiều hơn đi với nhóm	Hết chỗ, tốn thời gian tìm chỗ, đến thư viện hơi tối ở khu Sofa, thư viện ồn không tập trung được, mạp khó xài và không thực sự thể hiện hết các chức năng, khó tìm các sách mình mong muốn	Có	Vấn đề tìm sách, do cảm thấy vấn đề tìm chỗ khó giải quyết hơn	Có tầm 5-10p, chỗ ngồi thì để tìm hơn, chứ mà tìm sách thì lâu, đôi khi phải nhờ đến sự trợ giúp từ thủ thư	Đông người, nhưng mà không khí tĩnh lặng lắm, ai cũng chạy đl của mọi người	Có thính thoảng hơi ồn, nhưng hầu hết là yên tĩnh lắm	Muốn kiếm được sách để hơn chứ đi hoài mà tìm vẫn không ra	Hỏi thủ thư, hoặc mua ngoài luôn nếu có bán	Nói chuyện với thủ thư hỏi vị trí	Nhiều lúc mệt chỉ mu chỗ để ngồi hoặc nằm ngon chứ không muốn sách nữa
K50	Nam	Có	3-4/tuần	Chạy deadline, làm bài nhóm	Tuỳ lúc	Việc tìm sách ở thư viện quá phức tạp	Có	Vấn đề tìm sách	Có, không rõ thời gian cụ thể, chỉ nhớ đi cũng nhiều vòng lắm	Đông lắm, mà mọi người làm việc của mọi người đó khu quiet study im lặng lắm	Không, im lặng lắm, có khi rớt cây kim còn nghe tiếng	Khá khó chịu, kiểu cảm giác mọi người nhìn mình	Hỏi thủ thư, hoặc không tìm được thì đi về	Hỏi thủ thư, có đi vợt bạn thì nhờ bạn cùng kiếm	Nhiều người nhìn
K49	Nam	Có	>2/ trên tuần	Đọc sách, tận hưởng cảnh đẹp, có không gian tự học	Tùy lúc	Hết chỗ, không biết chỗ nào còn bàn, mọi người đi đó ở bàn học và bỏ đi	Có	Vấn đề hết chỗ, việc để giữ chỗ, tìm chỗ	Có, tầm 7 phút	khá đông, hết bàn, các khu vực đều kín	không ồn lắm	Thấy hơi chán, tốn công sức lên thư viện	Cafe	không	không
K49	Nữ	Có	3-4/tuần	Học bài, làm bt nhóm	Tuỳ lúc	Kệ sách ở thư viện xếp các đầu sách rồi khó hình dung được vị trí chính xác của sách mà tôi muốn tìm	Có	Vị trí chính xác khi tìm sách	Khá lâu nha, tại không hình dung được vị trí, đôi khi nhìn không rõ tên sách là bị hướt qua, phải tìm lại lần nữa	Nhiều người, mà lâu lâu mọi người hay nhìn	Không để ý lắm, nhưng do khu đó là khu quiet nên cũng không ồn đây	Bị bực do tìm hoài không thấy sách	Đi về, không muốn tìm nữa hôm	Hỏi thủ thư	Chắc là bị mệt đi, tìm để về
K49	Nam	thỉnh thoảng	1 ngày/tuần	làm bài	Một mình	Hết chỗ. hết ổ điện. đi kiếm chỗ. hone biết	có	Tìm chỗ. hết chỗ.	Có. tầm 3 nhút	Đông. khá bất tiện	Không. do nước trong	khá bất tiện	cafe	Không	Không

Trang tính1 | Formative Testing

Interview participants – primarily students who frequently visit the UEH library – reported that the current book-finding process involves multiple complex steps and is time-consuming. Although the system has divided the collection into areas and labeled each shelf by subject, many still spend 5–10 minutes moving between floors to locate the correct spot. Some students said they must stop to ask library staff or enlist friends' help to be sure of the location, especially when books have been temporarily moved to the loan-processing area. Furthermore, the online search tool only returns the shelf code and floor without showing book images or a detailed map, and it does not provide directions or alerts when a book is not in its expected location.

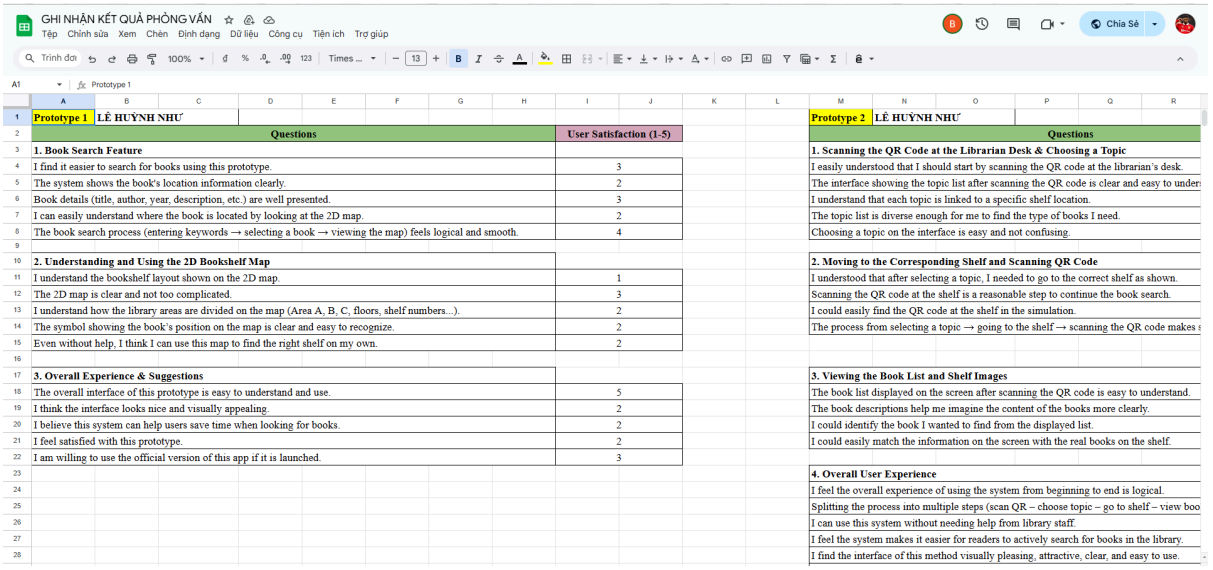
In addition, our team spent time asking about the library's shelving rules to better understand students' difficulties and to ground our solutions in the library's actual practices. The information we gathered from library staff includes:

+ Books are usually shelved according to their publication year.
+ Each day, staff spend approximately 30–60 minutes reshelving returned books.
+ Reshelving typically takes place near the library's closing hours for a final check.

+ On average, 5–10 books are borrowed per day, and most students using the book-locator machine still require staff assistance.
+ There are occasions when the system shows a book as available but it cannot be found on the shelves.
+ Staff hope for improvements that will help students visualize where books are located, as it can be difficult to imagine shelf positions.
+ They also wish to be able to adjust shelf height—avoiding shelves that are too high or too low—because some students' limited height makes locating books more challenging.

→ For a comprehensive overview of all user interview findings, please see the consolidated results here: Link

5.1.2. Lo-fi testing



GHI NHẬN KẾT QUẢ PHỎNG VẤN

Prototype 1	LÊ HUỲNH NHƯ	
	Questions	User Satisfaction (1-5)
1. Book Search Feature		
I find it easier to search for books using this prototype.		3
The system shows the book's location information clearly.		2
Book details (title, author, year, description, etc.) are well presented.		3
I can easily understand where the book is located by looking at the 2D map.		2
The book search process (entering keywords → selecting a book → viewing the map) feels logical and smooth.		4
2. Understanding and Using the 2D Bookshelf Map		
I understand the bookshelf layout shown on the 2D map.		1
The 2D map is clear and not too complicated.		3
I understand how the library areas are divided on the map (Area A, B, C, floors, shelf numbers...).		2
The symbol showing the book's position on the map is clear and easy to recognize.		2
Even without help, I think I can use this map to find the right shelf on my own.		2
3. Overall Experience & Suggestions		
The overall interface of this prototype is easy to understand and use.		5
I think the interface looks nice and visually appealing.		2
I believe this system can help users save time when looking for books.		2
I feel satisfied with this prototype.		2
I am willing to use the official version of this app if it is launched.		3

Prototype 2	LÊ HUỲNH NHƯ
	Questions
1. Scanning the QR Code at the Librarian Desk & Choosing a Topic	
I easily understood that I should start by scanning the QR code at the librarian's desk.	
The interface showing the topic list after scanning the QR code is clear and easy to under	
I understand that each topic is linked to a specific shelf location.	
The topic list is diverse enough for me to find the type of books I need.	
Choosing a topic on the interface is easy and not confusing.	
2. Moving to the Corresponding Shelf and Scanning QR Code	
I understood that after selecting a topic, I needed to go to the correct shelf as shown.	
Scanning the QR code at the shelf is a reasonable step to continue the book search.	
I could easily find the QR code at the shelf in the simulation.	
The process from selecting a topic → going to the shelf → scanning the QR code makes	
3. Viewing the Book List and Shelf Images	
The book list displayed on the screen after scanning the QR code is easy to understand.	
The book descriptions help me imagine the content of the books more clearly.	
I could identify the book I wanted to find from the displayed list.	
I could easily match the information on the screen with the real books on the shelf.	
4. Overall User Experience	
I feel the overall experience of using the system from beginning to end is logical.	
Splitting the process into multiple steps (scan QR – choose topic – go to shelf – view boo	
I can use this system without needing help from library staff.	
I feel the system makes it easier for readers to actively search for books in the library.	
I find the interface of this method visually pleasing, attractive, clear, and easy to use.	

The result tables and the video demo test will be fully compiled presented the link provided: Result table;

5.1.3. Hi-fi testing

To ensure that qualitative feedback was analyzed systematically, our team organized user interview responses into a structured **Interview Insights Summary Table**, aligned with the **eight core HCI principles**: Usability, Learnability, Efficiency, Satisfaction, Mental Model, Visibility & Feedback, Error Prevention & Recovery, and Accessibility. Each row of the table includes:

- **Key Insight** – A summary of recurring user feedback
- **Representative Quote** – One or more user quotes illustrating that insight

- **Recommendation** – A specific design suggestion based on the finding

HCI Aspect	Key Insight	Representative Quote	Recommendation
Usability	All users found the system easy to use without guidance.	"Dễ sử dụng, dễ hiểu." – User 2	No action needed for usability.
Learnability	Users were able to figure out how to use the system quickly (within 3 minutes).	"Nếu tự mò thì không quá 3 phút." – Khánh	Optional: Add a short onboarding guide for non-tech users.
Efficiency	Users reported faster task completion compared to the UEH library website.	"Nhanh, tiện, có thể xem ở nhà." – Khánh	No change needed, but maintain lightweight performance.
Satisfaction	Users were highly satisfied, scoring 8.5–9.5/10.	"Cho các bạn 8.5/10 đến 9.5/10." – Users 2, Khánh, Khôi	Continue reinforcing positive UX features and clarity.
Mental Model	Most users found the flow logical; one user experienced confusion.	"Không giống như mong đợi." – User 2 / "hong rõ." – User 3	Add visual flow hints or first-time-use tooltips.
Visibility & Feedback	System feedback was consistently described as smooth and clear.	"Mượt, rõ ràng, dễ hiểu." – Khánh, User 2, Khôi	Maintain clear visual feedback and transitions.
Error Prevention &	No critical errors	"Không có thao tác	Clarify library selection

Recovery	occurred; one prototype flaw may cause confusion.	sai." – Khánh / "Prototype chưa hoàn chỉnh." – User 2	UI to prevent misclicks.
Accessibility	Visually accessible for most, but potential issues for low-vision users.	"Thị giác ổn, nhưng người thị lực kém có thể gặp khó khăn." – User 2	Add accessibility options: larger text, contrast mode.

Table 3: *Summary of User Feedback Based on HCI Aspects*

→ Below is the list of references used in this document, including: RESULT, VIDEO/PICTURES

5.2. Quantitative Results

5.2.1. Lo-fi testing

Prototype	Aspect	Lê Huỳnh Như	Châu Quỳnh Như	Vũ Nguyễn Minh Quân	Lưu Phan Bình Thy
Prototype 1	*Book Search Feature*	2.8	2.8	3.2	3.2
	Understanding and Using the 2D Bookshelf Map	2	2.4	2.6	2.6
	Overall Experience & Suggestions	2.8	2.6	3	3
Prototype 2	*Scanning the QR Code at the Librarian Desk & Choosing a Topic*	3.2	3.2	3.4	3.2

	Moving to the Corresponding Shelf and Scanning QR Code	2	3.75	2.75	1.75
	Viewing the Book List and Shelf Images	3.75	3.5	3.25	3.75
	Overall User Experience	2.5	3.67	2.83	2.16
Prototype 3	*Ease of Use & Learnability*	4	4.2	4.2	2.8
	Library Map Feature	4.3	4.3	4.67	4.3
	Book Search Feature	4.5	4.25	4.75	4
	Overall User Experience	3.75	3.25	4.25	3.75

Across three prototypes evaluated by 4 students (Lê Huỳnh Như, Châu Quỳnh Như, Vũ Nguyễn Minh Quân, and Lưu Phan Bình Thy), Prototype 1 recorded relatively low average scores on all aspects: Book Search Feature **(2.95),** Understanding and Using the 2D Bookshelf Map **(2.4),** and Overall Experience **(2.85).** Prototype 2 showed marked improvement in Scanning the QR Code at the Librarian Desk & Choosing a Topic **(3.3)** and Viewing the Book List and Shelf Images **(3.56)**, but its Moving to the Corresponding Shelf and Scanning QR Code aspect fluctuated **(2.56)**, resulting in an overall score of **2.79**. Notably, Prototype 3 achieved the highest scores for Ease of Use & Learnability **(4.05)**, Library Map Feature **(4.43)**, and Book Search Feature **(4.38),** culminating in an Overall User Experience of **3.88**. The consistency among participants and the increasing satisfaction across versions highlight a clear direction: **Prototype 3: Prioritizing an interactive map and an intuitive search interface.**

5.2.2. Hi-fi testing

As our scale ranges from 1 to 5, with 3 representing the point where the proposed system is equally as good as the old system, any score greater than or equal to 3 can be interpreted as an indication of how much better the proposed system is

compared to the old one. The survey results indicate that all participants gave positive feedback on the proposed system, with average scores ranging from **3.13 to 4.75** on a 1–5 scale. Notably, criteria such as "ease of use," "user satisfaction," and "system responsiveness" received very high scores (above 4.5).

Based on this data, it is clear that **the proposed system outperforms the traditional system** in terms of efficiency, user-friendliness, and overall experience. This conclusion is strongly supported by the descriptive data, without the need for hypothesis testing. The table below summarizes the average scores for each HCI-related aspect:

HCI Aspect	Q1 Avg	Q2 Avg	Aspect Avg
Usability	4.75	4.75	4.75
Learnability	4.25	3.75	4
Efficiency	4	4.5	4.25
Satisfaction	4	4.5	4.25
Mental Model	3.5	3.75	3.63
Visibility & Feedback	4.75	3.25	4
Error Prevention	4	3.5	3.75
Accessibility	3.25	4	3.63

Table 4: *Summary Table of Average Scores (by HCI Aspect)*

5.3. Qualitative Insights

5.3.1. Lo-fi testing

After analyzing the results of the Formative Testing, combined with users' descriptive feedback about their experience, we identified several key points:

- Most users agreed that **Prototype 3** is the best version and provides the most outstanding and unique experience among the three prototypes.
- The **library map** and **visual display of books** in Prototype 3 helped users find the books they needed **much more easily**.
- The interface of Prototype 3 includes **many features**, but it is **not very user-friendly** and can feel a bit **confusing** at first glance.
- In Prototype 3, the **basic information of the library** is fully shown. However, the fact that **users can search for books and view library info without logging in** raises **security concerns**.
- Although the **book searching process** in Prototype 3 is excellent and helps users find books very easily, **one user mentioned** that the process in Prototype 2 might allow for **faster book searching** — which is important, since this task should not take too much time in a library.

HCI Aspect	**Insights**
(Usability)	In Prototype 3, on the initial interface, several buttons representing functions not directly related to the book-searching process were also included, making it difficult for users to determine where to start. The map feature in Prototype 3, although detailed, also caused some confusion about where users should begin interacting. While Prototype 3 provides an easier book-borrowing process, Prototype 2 has a more streamlined flow with fewer steps, which could make book searching faster.
(Interaction metaphor)	Prototype 3 lacks instructional text or guiding symbols to help users know where to start interacting.
(Accessibility)	**Feedback Insight:** - Users indicated that although the system provides a visual map, it lacks clear guided instructions. This makes it challenging for students who are unfamiliar with the library layout. - Additionally, the absence of assistive features, such as voice guidance or text enlargement, limits accessibility for users with visual or cognitive impairments.

	Improvement Suggestions: - Integrate a guided feature that offers step-by-step instructions to navigate the library. - Include accessibility settings such as text resizing, voice assistance, and color contrast adjustments. - Provide interactive prompts and tooltips to guide users through the system.
(Mental Model)	**Feedback Insight:** - The placement of buttons and the organization of features do not align with users' natural expectations. This misalignment makes it difficult for users to intuitively understand the next step. - Users expressed uncertainty about the camera's detection process, questioning whether it functions in real time or not. This lack of clarity disrupts the expected mental flow of the task. **Improvement Suggestions:** - Reorganize buttons to follow a more logical and sequential order, aligned with the typical user journey. - Clearly indicate when the camera is actively detecting, using visual cues like a blinking icon or status message. - Conduct usability testing specifically focused on button placement and real-time feedback to validate the redesign.

5.3.2. Hi-fi testing

- **User-friendly and accessible interface:** Most users stated that the interface is intuitive, easy to understand, and aligned with their technology usage habits. They were able to use the system smoothly without any initial instructions. This indicates that the current interface design meets expectations in terms of usability.
- **Lack of clear guidance in user flow:** Some users expressed confusion in identifying the next steps during usage. The absence of current-position indicators (e.g., breadcrumbs or step-by-step guides) made them feel disoriented and led to trial-and-error attempts to complete tasks.
- **Unclear error correction reduces trust:** When users made mistakes, many were unsure how to undo or fix them. Some feedback mentioned the lack of

warnings when actions failed. This created anxiety and lowered user trust in the system's reliability.

- **Text display not suitable for all audiences:** One suggestion was that the font size in the book detail section is too small, making it difficult for visually impaired or elderly users. This highlights the need for better accessibility to accommodate a wider range of users.
- **Fast and reliable system feedback:** Most users were satisfied with how the system responded after each interaction. It clearly showed whether an action was processed, completed, or failed, making users feel informed and confident.
- **High satisfaction and interest in real-world implementation:** Many participants stated that they would use the system if it were implemented at UEH's library. This demonstrates that the system not only functions well but also delivers a compelling experience that meets real student needs.

Key Strengths	**Areas for Improvement**
Simple and easy-to-use interface	Lack of clear guidance in user flow
Fast and smooth system feedback	No clear error-handling mechanism
Positive user experience	Interface not yet suitable for users with low vision
High potential for real-world adoption	Lacks onboarding or initial instructions for new users

Table 5: *Summary of Strengths and Improvement Areas from User Evaluation*

→ **Conclusion:** The UEH Library's book search and navigation system was highly rated by users across several criteria, particularly in terms of interface design, response speed, and convenience. However, for the system to be widely adopted and effective in real-world use, the development team should improve user flow guidance, error recovery mechanisms, and accessibility support. If these aspects are addressed, the system has strong potential to become an effective digital transformation solution for university libraries.

CHAPTER 6: CONCLUSION

6.1. Conclusion

6.1.1. Strengths

The system is designed to be user-friendly and easy to use, helping UEH students search for and borrow books more quickly. The interface adopts similar colors and layout to the existing UEH Library application, ensuring consistency in user experience and allowing users to easily get familiar with the system from their first use. In addition, the integration of a virtual library map combined with book-locating cameras enables users to visually and efficiently identify the location of the materials they need, thereby reducing search time and improving usage effectiveness.

6.1.2. Weakness

Although the system has many strengths, some design weaknesses remain. Users sometimes struggled to know the next step due to unclear instructions or lack of indicators. When errors occurred, like entering incorrect information, the system didn't provide clear messages or guidance, confusing. The book detail section also wasn't optimized for users with visual impairments, using small fonts and low contrast. Due to time and resource limits, not all intended features were implemented. These issues should be addressed in future improvements to enhance the overall user experience.

6.1.3. Limitations

Due to time limitations, the team was only able to conduct testing with four participants, all of whom were students already familiar with the UEH Library. As a result, the feedback collected lacked diversity. In addition, the virtual 3D library map is currently only presented as an illustration and has not yet been integrated into the system. The prototype focuses solely on UEH B Library, while several other features, such as advanced search, accessibility support, and expansion to other libraries, could not be implemented due to limited time and resources.

6.2. Future Developments

To enhance the system and make it more user-friendly, the following future developments are recommended:

+ Add clearer user guidance, such as step-by-step instructions.
+ Improve error messages to help users understand what went wrong and how to fix it.
+ Provide adjustable font size options to improve accessibility for users with visual impairments.
+ Enhance the smart search function by adding keyword suggestions, filters, and personalized search results.
+ Integrate a complete 3D virtual map of UEH Library at both Campus B and Campus N, along with navigation buttons.
  Improve visibility of user progress with features like breadcrumbs and "You are here" indicators.

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